

640:481 WS17 Austin DeCicco

$$1, T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad P(-t_{\alpha/2, n-1} < T < t_{\alpha/2, n-1}) = 1 - \alpha$$

Let $z = t_{\alpha/2, n-1}$, then $P(-z < T < z) = 1 - \alpha$

$$P(-z < T < z) = P\left(-z < \frac{\bar{X} - \mu}{S/\sqrt{n}} < z\right) = P\left(-z \frac{S}{\sqrt{n}} < \bar{X} - \mu < z \frac{S}{\sqrt{n}}\right) =$$

$$P(-z \frac{S}{\sqrt{n}} - \bar{X} < -\mu < z \frac{S}{\sqrt{n}} - \bar{X}) = P(\bar{X} - z \frac{S}{\sqrt{n}} < \mu < \bar{X} + z \frac{S}{\sqrt{n}}) =$$

$$P(\bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}) = 1 - \alpha \quad \square$$